DROUGHTS

DEFINITION

DROUGHT-
is an insidious natural hazard that results from a departure of precipitation from expected or normal that, when extended over a season or longer period of time, is insufficient to meet the demands of human, plant and animal activities.

ONSET TYPE & WARNING SYSTEM

ONSET TYPE AND WARNING

Drought is a **slow-onset** disaster and as it is difficult to demarcate the time of its onset and end. Falling rainfall levels, falling groundwater levels, drying wells, rivers and reservoirs, and poor agricultural production warn the onset of drought. According to the Indian Meteorological Department, the country is said to be drought affected when the overall **rainfall deficiency** is more than 10 per cent of the long period average and more than 20 per cent of the country area is affected by such drought conditions.

ELEMENTS AT RISK

Drought impacts mostly rainfed crops to start with and subsequently the irrigated crops. Areas with minimum of alternative water sources to rainfall (ground and canal water supplies), areas subjected to drastic environmental degradation such as denuded forest lands and altered ecosystems, and areas where livelihoods alternative to agriculture are least developed are most vulnerable to drought. The herdsman, landless laborers, subsistence farmers, the women, children, and farm animals are the most vulnerable groups affected by the drought conditions.

Precipitation
Groundwater
Reservoirs
Canals
Soil moisture
Ponds/tanks

Agriculture
Domestic
Industry
Natural vegetation
Recreation
Wild life

A simple way of finding drought vulnerability of a region is to calculate the balance between different sources and users of water.

Different Types of Drought:

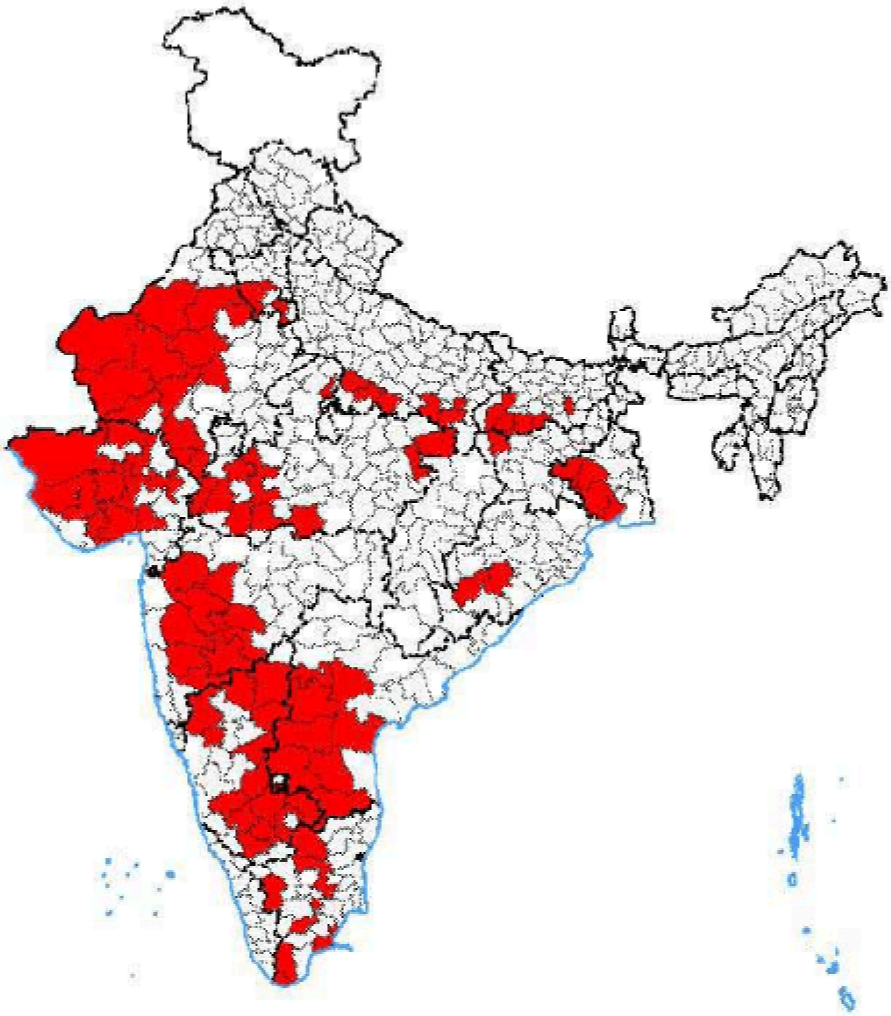
1. Meteorological Drought:is based on the degree of dryness or rainfall deficit and the length of the dry period.

2. **Hydrological Drought**: is based on the impact of rainfall deficits on the water supply such as stream flow, reservoir and lake levels, and ground water table decline.

3. **Agricultural Drought:**refers to the impacts on agriculture by factors such as rainfall deficits, soil water deficits, reduced ground water, or reservoir levels needed for irrigation.

4. **Socio-Economic Drought:** considers the impact of drought conditions (meteorological, agricultural, or hydrological drought) on supply and demand of some economic goods such as

HAZARD ZONES



Map of drought prone districts of India (Drought prone areas are in red).

INDIAN CONTEXT :

In the past, droughts have periodically led to major Indian famines, including the Bengal famine of 1770, in which up to one third of the population in affected areas died; the 1876–1877 famine, in which over five million people died; and the 1899 famine, in which over 4.5 million died.[3][4] In simple words, drought has destroyed India on a large scale. Eighteen meteorological and 16 hydrological droughts occurred in India between 1870 and 2018. The most severe meteorological droughts were in the years 1876, 1899, 1918, 1965, and 2000, while the five worst hydrological droughts occurred in the years 1876, 1899, 1918, 1965, and 2000. The drought of 1899 can be classified as meteorological as well as hydrological and was the most severe documented drought India has ever experienced to date.

CAUSES :

A **drought** is an event of prolonged shortages in the water supply, whether atmospheric (below-average precipitation), surface water or ground water. A drought can last for months or years, or may be declared after as few as 15 days. It can have a substantial impact on the ecosystem and agriculture of the affected region and harm to the local economy. Annual dry seasons in the tropics significantly increase the chances of a drought developing and subsequent bush fires. Periods of heat can significantly worsen drought conditions by hastening evaporation of water vapour.

In El Niño years, India often faces warmer temperatures and less rainfall, causing droughts in some regions. This affects agriculture, water resources, and ecosystems.

Causes of Drought

- Deficit rainfall in terms of time and space
- Over population
- Over grazing
- Deforestation
- Soil erosion
- Excessive use of ground and surface water
- Loss of biodiversity
- Low Moisture holding capacity
- Absence of irrigation facilities
- Livestock without adequate fodder
- storage facilities
- Poor water management
- Water consuming cropping patterns
- Urbanization
- Industrialization
- Global warming

Criteria for Drought estimation

The following criteria have been set by the Indian Meteorological Division (IMD) for identifying the drought.

Onset of drought: Deficiency of a particular year's rainfall exceeding 25 per cent of normal.

Moderate drought: Deficit of rainfall between 26-50 per cent of normal.

Severe drought: Deficit of rainfall more than 50 per cent of normal.

- Around 68 per cent of India's total area is drought prone to drought.
- 35% receives rainfall of 750 to 1150mm which is considered to be drought prone.
- 33% receives rainfall of less than 750mm which is considered to be chronically drought prone.
- 315 out of a total of 725 Talukas in 99 districts are drought prone.
- 50 million people are annually affected by drought.
- In 2001 more than eight states suffered the impact of severe drought.
- In 2003 most parts of Rajasthan experienced the fourth consecutive year of drought.

TYPICAL EFFECTS

Drought, different from other natural disasters, do not cause any structural damages. The typical effects include loss of crop, dairy, timber (forest fires), and fishery production; increase in energy demand for pumping water; reduced energy production; increased unemployment, loss of biodiversity, reduced water, air, and landscape quality; groundwater depletion, food shortage, health reduction and loss of life, increased poverty, reduced quality of life, and social unrest leading to migration.

It is a natural disaster, which is hazardous to human beings because it results in water shortage, damages to crops, and an increased death rate of livestock and wild animals.

It also results in shortage of electricity. Reports show, many people worldwide die during the these extreme conditions.

In drought-prone areas certain measures such as construction of reservoirs, rain-harvest system and stopping over-grazing could be taken. It causes increase in food prices and unemployment.

Impacts of Drought:

Environmental:

- Moisture Stress
- Drinking Water Shortage
- Damage to Natural Vegetation and Various Ecosystems
- Increased Air And Water Pollution

Socio-economic:

- Malnutrition
- Poor Hygiene
- Ill Health
- Migration
- Increased Stress and Morbidity
- Social Strife

Before a Drought

The best way to prepare for a drought is to conserve water. Make conserving water a part of your daily life.

Indoor Water Conservation Tips Before a Drought

GENERAL

- Never pour water down the drain when there may be another use for it. For example, use it to water your indoor plants or garden.
- Fix dripping faucets by replacing washers. One drop per second wastes 2,700 gallons of water a year.
- Check all plumbing for leaks and have any leaks repaired by a plumber.
- Retrofit all household faucets by installing aerators with flow restrictors.
- Install an instant hot water heater on your sink.
- Insulate your water pipes to reduce heat loss and prevent them from breaking.
- Install a water-softening system only when the minerals in the water would damage your pipes. Turn the softener off while on vacation.
- Choose appliances that are more energy and water efficient.

BATHROOM

- Consider purchasing a low-volume toilet that uses less than half the water of older models. Note: In many areas, low-volume units are required by law.
- Install a toilet displacement device to cut down on the amount of water needed to flush. Place a one-gallon plastic jug of water into the tank to displace toilet flow. Make sure it does not interfere with the operating parts.
- Replace your showerhead with an ultra-low-flow version.

KITCHEN

- Instead of using the garbage disposal, throw food in the garbage or start a compost pile to dispose it.

Outdoor Water Conservation Tips Before a Drought

GENERAL

- Check your well pump periodically. If the automatic pump turns on and off while water is not being used, you have a leak.
- Plant native and/or drought-tolerant grasses, ground covers, shrubs and trees. Once established, your plants won't need as much watering. Group plants together based on similar water needs.
- Don't buy water toys that require a constant stream of water.
- Don't install ornamental water features (such as fountains) unless they use re-circulated water.
- Consider rainwater harvesting where practical.
- Contact your local water provider for information and assistance.

LAWN CARE

- Position sprinklers so water lands on the lawn and shrubs and not on paved areas.
- Repair sprinklers that spray a fine mist.
- Check sprinkler systems and timing devices regularly to be sure they operate properly.
- Raise the lawn mower blade to at least three inches or to its highest level. A higher cut encourages grass roots to grow deeper and holds soil moisture.
- Plant drought-resistant lawn seed. Reduce or eliminate lawn areas that are not used frequently.
- Don't over-fertilize your lawn. Applying fertilizer increases the need for water. Apply fertilizers that contain slow-release, water-insoluble forms of nitrogen.
- Choose a water-efficient irrigation system such as drip irrigation for your trees, shrubs and flowers.
- Turn irrigation down in fall and off in winter. Water manually in winter only if needed.
- Use mulch around trees and plants to retain moisture in the soil. Mulch also helps control weeds that compete with plants for water.
- Invest in a weather-based irrigation controller—or a smart controller. These devices will automatically adjust the watering time and frequency based on soil moisture, rain, wind, and evaporation and transpiration rates. Check with your local water agency to see if there is a rebate available for the purchase of a smart controller.

POOL

- Install a new water-saving pool filter. A single back flushing with a traditional filter uses 180 to 250 gallons of water.
- Cover pools and spas to reduce water evaporation.

Indoor Water Conservation Tips During a Drought

BATHROOM

- Avoid flushing the toilet unnecessarily. Dispose of tissues, insects and other similar waste in the trash rather than the toilet.
- Take short showers instead of baths. Turn on the water only to get wet and lather and then again to rinse off.
- Avoid letting the water run while brushing your teeth, washing your face or shaving.
- Place a bucket in the shower to catch excess water for watering plants.

KITCHEN

- Operate automatic dishwashers only when they are fully loaded. Use the "light wash" feature to use less water.
- Hand wash dishes by filling two containers—one with soapy water and the other with rinse water containing a small amount of chlorine bleach.
- Clean vegetables in a pan filled with water rather than running water from the tap.
- Store drinking water in the refrigerator. Do not let the tap run while you are waiting for water to cool.
- Avoid wasting water waiting for it to get hot. Capture it for other uses such as plant watering or heat it on the stove or in a microwave.
- Don't rinse dishes before placing them in the dishwasher, just remove large particles of food.
- Avoid using running water to thaw meat or other frozen foods. Defrost food overnight in the refrigerator or use the defrost setting on your microwave.

LAUNDRY

- Operate clothes washers only when they are fully loaded or set the water level for the size of your load.

Outdoor Water Conservation Tips During a Drought

CAR WASHING

- Use a commercial car wash that recycles water.
- If you wash your own car, use a shut-off nozzle that can be adjusted down to a fine spray on your hose.

LAWN CARE

- Avoid over watering your lawn and water only when needed.
- A heavy rain eliminates the need for watering for up to two weeks. Most of the year, lawns only need one inch of water per week.

- Check the soil moisture levels with a soil probe, spade or large screwdriver. You don't need to water if the soil is still moist. If your grass springs back when you step on it, it doesn't need water yet.
- If your lawn does require watering, do so early in the morning or later in the evening, when temperatures are cooler.
- Check your sprinkler system frequently and adjust sprinklers so only your lawn is watered and not the house, sidewalk, or street.
- Water in several short sessions rather than one long one, in order for your lawn to better absorb moisture and avoid runoff.
- Use a broom or blower instead of a hose to clean leaves and other debris from your driveway or sidewalk.
- Avoid leaving sprinklers or hoses unattended. A garden hose can pour out 600 gallons or more in only a few hours.
- In extreme drought, allow lawns to die in favor of preserving trees and large shrubs.

MAIN MITIGATION STRATEGIES

Drought monitoring is continuous observation of rainfall situation, water availability in reservoirs, lakes, rivers and comparing with the existing water needs of various sectors of the society.

Rooftop rainwater harvesting: Roof acts as catchment draining the water to a cistern.

Water supply augmentation and conservation through rainwater harvesting in houses and farmers' fields increases the content of water available. *Water harvesting* by either allowing the runoff water from all the fields to a common point (e.g. Farm ponds, see the picture) or allowing it to infiltrate into the soil where it has fallen (*in situ*) (e.g. contour bunds, contour cultivation, raised bed planting etc) helps increase water availability for sustained agricultural production.

Expansion of **irrigation** facilities reduces the drought vulnerability. **Land use** based on its capability helps in optimum use of land and water and can avoid the undue demand created due to their misuse.

Farm ponds collected the rainwater and helped in escaping the drought impacts in Karnool District of Andhra Pradesh.

Livelihood planning identifies those livelihoods which are least affected by the drought. Some of such livelihoods include increased off-farm employment opportunities, collection of non-timber forest produce from the community forests, raising goats, and carpentry etc.

Drought Prevention and Mitigation:

Drought can be mitigated by two kinds of measures, either by adopting preventive measures or by developing a preparedness plan

Table 7. Preventive measures and preparedness plan for drought mitigation

Preventive measures	Preparedness plan
• Dams/reservoirs and wetlands to store water • Watershed management • Water rationing • Cattle management • Proper selection of crop for drought-affected areas • Levelling, soil-conservation techniques • Reducing deforestation and fire-wood cutting in the affected areas • Alternative land-use models for water sustainability • Checking of migration and providing alternate employment • Education and training to the people • Participatory community programmes	• Improvement in agriculture through modifying cropping patterns and introducing drought-resistant varieties of crops • Management of rangeland with improvement of grazing patterns, introduction of feed and protection of shrubs and trees. • Development of water resource system with improved irrigation, development of improved storage facilities, protection of surface water from evaporation and introduction of drop irrigation system • Animal husbandry activities can help in mitigation with use of improved and scientific methods

COMMUNITY BASED MITIGATION

Drought planning

The basic goal of drought planning is to improve the effectiveness of preparedness and response efforts by enhancing monitoring, mitigation and response measures. Planning would help in effective coordination among state and national agencies in dealing with the drought. Components of drought plan include establishing drought taskforce which is a team of specialists who can advise the government in taking decision to deal with drought situation, establishing coordination mechanism among various agencies which deal with the droughts, providing crop insurance schemes to the farmers to cope with the drought related crop losses, and public awareness generation.

Public awareness and education

Educating the masses on various strategies you learned above would help in effective drought mitigation. This includes organizing drought information meetings for the public and media, implementing water conservation awareness programs in the mass media like television, publishing and distributing pamphlets on water conservation techniques and agricultural drought management strategies like crop contingency plans and rainwater harvesting and establishing drought information centers for easy access to the farmers.

A watershed showing collection of water to common point. *ImageSource: http://www.4j.lane.edu*

Watersheds: For water supply augmentation & conservation

Watersheds are the geographic areas where the water flows to a common point. To mitigate the drought impact, all kinds of soil and water conservation measures are taken up with the involvement of the local communities. This approach helped these areas to manage efficiently the soil, vegetation, water and other resources. By conserving scarce water sources and improving the management of soil and vegetation, watersheds have the potential to create conditions conducive to higher agricultural productivity while conserving natural resources.

Checkdams (Bhanadaras)

These are check dams or diversion weirs built across rivers. A traditional system found in Maharashtra, their presence raises the water level of the rivers so that it begins to flow into channels. They are also used to impound water and form a large reservoir. Where a bandhara was built across a small stream, the water supply would usually last for a few months after the rains.

Source: http://www.rainwaterharvesting.org

What a mitigation approach can do? A success story

Ralegan, before drought mitigation efforts

Ralegan, after drought mitigation efforts

The people of Ralegan Siddhi in Maharashtra transformed the dire straits to prosperity. Twenty years ago the village showed all traits of abject poverty. It practically had no trees, the topsoil had blown off, there was no agriculture and people were jobless. Anna Hazare, one of the India's most noted social activists, started his movement concentrating on trapping every drop of rain, which is basically a *drought mitigation* practice.

So the villagers built check dams and tanks. To conserve soil they planted trees. The result: from 80 acres of irrigated area two decades ago, Ralegan Siddhi has a massive area of 1300 acres under irrigation. The migration for jobs has stopped and the per capita income has increased ten times from Rs.225 to 2250 in this span of time.

The entire effort was only people’s enterprise and involved no funds or support from the Government.